



Modelling Financial Markets

Chapter 4: Portfolio Optimisation



Daniel Traian PELE

Bucharest University of Economic Studies

IDA Institute Digital Assets

Blockchain Research Center

AI4EFin Artificial Intelligence for Energy Finance

Romanian Academy, Institute for Economic Forecasting

MSCA Digital Finance

Acronyms used in this material (1/2)

- ▣ AI: Artificial Intelligence
- ▣ BET: Bucharest Exchange Trading (index)
- ▣ BET-TR: BET Total Return (index)
- ▣ BVB: Bursa de Valori București (the Bucharest Stock Exchange)
- ▣ CAPM: Capital Asset Pricing Model
- ▣ DSR: Deflated Sharpe Ratio
- ▣ ERC: Equal Risk Contribution portfolio
- ▣ ETF: Exchange-Traded Fund
- ▣ FDR: False Discovery Rate
- ▣ FWER: Family-Wise Error Rate
- ▣ GLS: Generalised Least Squares
- ▣ GMV: Global Minimum-Variance portfolio
- ▣ GMV-LO: Global Minimum-Variance portfolio, Long-Only
- ▣ GMV-LW: Global Minimum-Variance portfolio with Ledoit–Wolf covariance
- ▣ HAC: Heteroskedasticity and Autocorrelation Consistent (standard errors)
- ▣ HRP: Hierarchical Risk Parity
- ▣ HYG: iShares iBoxx High Yield Corporate Bond ETF
- ▣ IEF: iShares 7–10 Year Treasury Bond ETF
- ▣ LO: Long-Only, no short sales
- ▣ LQD: iShares iBoxx Investment Grade Corporate Bond ETF
- ▣ ML: Machine Learning
- ▣ MV: Mean–Variance tangency portfolio estimated from sample moments
- ▣ MV-LO: Mean–Variance tangency portfolio, Long-Only
- ▣ NYSE: New York Stock Exchange
- ▣ OLS: Ordinary Least Squares
- ▣ OOS: Out-Of-Sample

Acronyms used in this material (2/2)

- POET: Principal Orthogonal compleMent Thresholding
- US: United States

Outline

Motivation

Mean–Variance Optimisation

The Limits of Markowitz

Covariance Shrinkage

Black–Litterman

Risk Parity and Hierarchical Risk Parity

Out-of-Sample Evaluation

A Portfolio of BVB Blue Chips

Case Study: The Implementable Efficient Frontier

AI for Scientific Discovery

Conclusions

References

Portfolio Optimisation: Easy to Write, Hard to Estimate

- A **portfolio** is a weight vector $w = (w_1, \dots, w_N)^\top$ with $w^\top \mathbf{1} = 1$; optimisation chooses w to maximise a criterion under constraints
- The inputs are **estimated**: N means and $N(N+1)/2$ variances and covariances
 - ▶ 16 assets: $16 + 136 = 152$ parameters from a 60-month window, i.e. 960 observations
- Uses: strategic allocation (pension funds), home bias of private investors, the covariance matrix of risk management (Chapters 6–7)
- Key message: judge every rule **out of sample**, after costs, with a test of the difference
- Textbook treatment: [Franke, Härdle & Hafner \(2019\)](#), Part I (portfolio theory)

The Central Tension

In theory

- ▣ Known means and covariances
- ▣ The mean–variance portfolio is optimal
- ▣ More assets always help

In practice

- ▣ Parameters are estimated with error
- ▣ The optimiser loads on the largest errors
- ▣ More assets, more parameters, more noise

- ▣ This chapter: better inputs (shrinkage, Black–Litterman), rules that ignore means (risk parity, HRP) and a fair out-of-sample comparison

Data Used in This Chapter

Universe	Assets	Period
Sectors	9 US sector ETFs (XLB ... XLY)	1999–2026
Multi-asset	SPY, EFA, EEM, TLT, IEF, LQD, HYG, GLD	2007–2026
Combined	9 sectors + 7 multi-asset ETFs (no SPY)	2007–2026
Romania	8 BVB blue chips, BET, BET-TR	2014–2026

- Monthly returns from month-end prices on common trading days; US data end in July 2026, BVB data in August 2026
- Risk-free rate: one-month Treasury bill, [Kenneth French Data Library](#)

The Assets

- ▣ Equities: SPY (S&P 500), EFA (developed markets outside North America), EEM (emerging markets)
- ▣ US Treasuries: TLT (maturities above 20 years), IEF (7–10 years)
- ▣ Credit: LQD (investment-grade corporate bonds), HYG (high-yield corporate bonds)
- ▣ Gold: GLD; sectors: XLK technology, XLU utilities, XLP consumer staples, . . .
- ▣ BVB: TLV, SNP, BRD, TGN, SNG, SNN, EL, TEL; BET is a price index, BET-TR reinvests dividends

Prerequisites Recap: Notation and Two Assets

- Simple return $R_{i,t} = P_{i,t}/P_{i,t-1} - 1$; excess return $R_{i,t}^e = R_{i,t} - R_{f,t}$; portfolio mean $w^\top \mu$, variance $w^\top \Sigma w$
- **Sharpe ratio** (Sharpe (1966)): $SR_p = \mathbb{E}[R_p^e]/\sigma(R_p^e)$; monthly figures annualised with $\times 12$ (mean) and $\times \sqrt{12}$ (volatility, Sharpe)
- **Maximum drawdown**: $\min_t (W_t / \max_{s \leq t} W_s - 1)$, the largest fall of wealth W_t from a previous peak
- Two assets, correlation ρ : $\sigma_p^2 = w^2 \sigma_1^2 + (1-w)^2 \sigma_2^2 + 2w(1-w)\rho\sigma_1\sigma_2$, minimised at $w^* = \frac{\sigma_2^2 - \rho\sigma_1\sigma_2}{\sigma_1^2 + \sigma_2^2 - 2\rho\sigma_1\sigma_2}$
- SPY and TLT, monthly 2002–2026: $\rho = -0.10$, $w^* = 46\%$ SPY, volatility 9.4% against 14.7% and 13.5% (time-varying correlation: Chapter 6)

The Markowitz Problem

- ▣ **Markowitz (1952)**: investors care about the mean and the variance of portfolio returns
- ▣ **Minimum-variance frontier**: minimum variance for every target mean m ; its upper branch, $m \geq \mu_{\text{GMV}}$ (the mean of the GMV portfolio), is the **efficient frontier**

$$\min_w w^\top \Sigma w \quad \text{s.t.} \quad w^\top \mu = m, \quad w^\top \mathbf{1} = 1$$

- ▣ Utility form, risk aversion $\gamma > 0$: $\max_w w^\top \mu - \frac{\gamma}{2} w^\top \Sigma w$ s.t. $w^\top \mathbf{1} = 1$; as γ varies it traces exactly the efficient branch
- ▣ Inputs: N means, N variances, $N(N-1)/2$ covariances
- ▣ Weights may be negative: a **short sale** (selling a borrowed asset)

Deriving the GMV and Tangency Portfolios for N Assets

- GMV: $\min_{\mathbf{w}} \mathbf{w}^\top \Sigma \mathbf{w}$ s.t. $\mathbf{w}^\top \mathbf{1} = 1$; Lagrangian $\mathcal{L} = \mathbf{w}^\top \Sigma \mathbf{w} - \lambda(\mathbf{w}^\top \mathbf{1} - 1)$
 - ▶ First-order condition $2\Sigma \mathbf{w} = \lambda \mathbf{1}$, so $\mathbf{w} = \frac{\lambda}{2} \Sigma^{-1} \mathbf{1}$; the constraint gives $\frac{\lambda}{2} = 1/A$, $A = \mathbf{1}^\top \Sigma^{-1} \mathbf{1}$
 - ▶ $\mathbf{w}_{\text{GMV}} = \Sigma^{-1} \mathbf{1}/A$ and $\sigma_{\text{GMV}}^2 = \mathbf{w}_{\text{GMV}}^\top \Sigma \mathbf{w}_{\text{GMV}} = 1/A$
- $N = 2$: $\Sigma^{-1} \propto \begin{pmatrix} \sigma_2^2 & -\rho\sigma_1\sigma_2 \\ -\rho\sigma_1\sigma_2 & \sigma_1^2 \end{pmatrix}$ returns $\mathbf{w}^*$ of the recap slide
- Tangency: maximise $\mathbf{w}^\top \boldsymbol{\mu} / \sqrt{\mathbf{w}^\top \Sigma \mathbf{w}}$; the condition $\boldsymbol{\mu} \propto \Sigma \mathbf{w}$ gives $\mathbf{w} \propto \Sigma^{-1} \boldsymbol{\mu}$
- Maximum squared Sharpe ratio: $\theta^2 = \boldsymbol{\mu}^\top \Sigma^{-1} \boldsymbol{\mu}$, the quantity estimated with bias in Section 3
- With Σ positive definite and $B = \mathbf{1}^\top \Sigma^{-1} \boldsymbol{\mu}$, the normalised weights $\Sigma^{-1} \boldsymbol{\mu} / B$ have Sharpe ratio $\text{sign}(B) \theta$
 - ▶ Maximum only if $B > 0$; $B < 0$ gives the minimum, $B = 0$ no fully invested solution

Tobin: Separation



James Tobin, Nobel Prize in Economics 1981

Photo: unknown author (1962); public domain;
Wikimedia Commons

- Tobin (1958): add a riskless asset paying R_f
- Every investor holds the **same** risky portfolio, mixed with cash or borrowing
- Risk aversion changes only the mix, not the risky portfolio: **two-fund separation**
- The best risky portfolio maximises the Sharpe ratio: the **tangency portfolio**
- Separation is the bridge from Markowitz to the CAPM (Chapter 3)

Two Special Portfolios

Global minimum variance (GMV)

- Lowest variance of all portfolios

$$w_{\text{GMV}} = \frac{\Sigma^{-1} \mathbf{1}}{\mathbf{1}^\top \Sigma^{-1} \mathbf{1}}$$

- Needs **no** expected returns

- In this chapter, MV denotes the tangency rule estimated from sample means and covariances

Tangency (maximum Sharpe)

- Highest Sharpe ratio

$$w_{\text{MV}} = \frac{\Sigma^{-1} \mu}{\mathbf{1}^\top \Sigma^{-1} \mu}$$

- μ : expected **excess** returns; requires $\mathbf{1}^\top \Sigma^{-1} \mu > 0$

Merton: the Frontier in Closed Form



Robert C. Merton, Nobel Prize in
Economics 1997

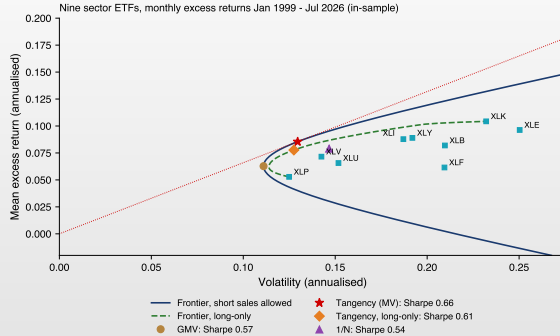
Photo: Digarnick (2006); public domain;
Wikimedia Commons

- Merton (1972): with short sales, the frontier is a hyperbola
- Let $A = \mathbf{1}^\top \Sigma^{-1} \mathbf{1}$, $B = \mathbf{1}^\top \Sigma^{-1} \boldsymbol{\mu}$, $C = \boldsymbol{\mu}^\top \Sigma^{-1} \boldsymbol{\mu}$, $D = AC - B^2$

$$\sigma^2(m) = \frac{Am^2 - 2Bm + C}{D}$$

- The GMV sits at $m = B/A$ with $\sigma^2 = 1/A$
- Every frontier portfolio is a mix of GMV and tangency

The Frontier of US Sectors



- Monthly excess returns, January 1999 – July 2026 ($T = 331$); all parameters estimated on the full sample

Reading the Sector Frontier

- ▣ Tangency: Sharpe 0.66; GMV: 0.57 (volatility 11.1%); $1/N$: 0.54
- ▣ Tangency weights: -0.37 financials, -0.21 materials, $+0.37$ health care
- ▣ Long-only tangency: Sharpe 0.61; materials, financials and industrials get exactly zero
 - ▶ A **corner solution**: the constraint binds and assets drop out
- ▣ The gain of the optimiser over $1/N$ is small even **in sample**, with perfect hindsight

Constraints

- ▣ **Long-only (LO)**: $w_i \geq 0$; no closed form, solved by quadratic programming
- ▣ Position limits: $w_i \leq 0.2$; sector or asset-class limits: $\sum_{i \in G} w_i \leq c_G$
- ▣ Leverage limit: $\sum_i |w_i| \leq L$
- ▣ [Frost & Savarino \(1988\)](#): constraining weights improved out-of-sample performance
- ▣ In this chapter: MV-LO and GMV-LO are the long-only versions of MV and GMV

Recap: Mean–Variance Optimisation

- ▣ The frontier trades mean against variance; correlation drives diversification
- ▣ Two special portfolios: GMV, which needs no means, and tangency, which maximises the Sharpe ratio
- ▣ With a riskless asset, the risky part of every efficient portfolio is the tangency portfolio
- ▣ Long-only constraints create corner solutions
- ▣ Everything so far assumed the parameters are **known**

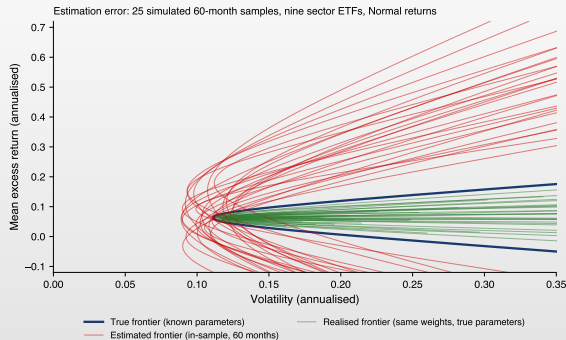
Means Are Hard to Estimate: Merton's Argument

- Merton (1980): Y years sampled h times a year, i.i.d. returns with annual mean μ and variance σ^2
 - ▶ $\text{Var}(\hat{\mu}) = \sigma^2/Y$: independent of h ; only the **span** of the sample helps
 - ▶ $\text{Var}(\hat{\sigma}^2) \approx 2\sigma^4/(hY) \rightarrow 0$ as $h \rightarrow \infty$: variances sharpen with frequency
- SPY, $Y = 24$, $\sigma = 14.7\%$: standard error 3.0%, 95% interval [5.8%, 17.6%] for a mean of 11.7%
- A standard error of 1% needs $(0.147/0.01)^2 \approx 217$ years
- Chopra & Ziemba (1993): errors in means cost about ten times more than errors in variances; rules without means (GMV, risk parity) are more robust

Error Maximisation

- Michaud (1989): mean–variance optimisers are **estimation-error maximisers**
 - ▶ They overweight assets with overestimated means and underestimated risk
- Best & Grauer (1991): tiny changes in a mean can move the weights of many assets
- In-sample performance is therefore **biased upwards**
- Experiment: simulate returns from known parameters, optimise, evaluate with the true parameters

Estimated versus Realised Frontiers



- True parameters: sector sample means and covariances, 1999–2026; returns drawn from the Normal distribution

Reading the Simulation

- ▣ Red: frontiers estimated from 60 months, far too optimistic
- ▣ Green: the same portfolios evaluated with the true parameters, all **below** the true frontier
- ▣ 2000 simulated tangency portfolios, 60 months each
 - ▶ Median estimated maximum Sharpe $\hat{\theta}$ 1.56; median true Sharpe 0.30; true optimum 0.66
 - ▶ In 11% of the samples $1^\top \hat{\Sigma}^{-1} \hat{\mu} < 0$: the normalised weights are then the in-sample minimum
 - ▶ 96% have a true Sharpe below $1/N$ (0.54); 7.5% below zero
- ▣ Even with Normal, stable returns, estimation error destroys the optimum

The Plug-in Sharpe Ratio Is Biased: an Exact Result

- i.i.d. $N(\boldsymbol{\mu}, \boldsymbol{\Sigma})$ returns, maximum-likelihood $\hat{\boldsymbol{\mu}}, \hat{\boldsymbol{\Sigma}}$: $\hat{\boldsymbol{\mu}} \sim N(\boldsymbol{\mu}, \boldsymbol{\Sigma}/T)$, $T\hat{\boldsymbol{\Sigma}} \sim W_N(T-1, \boldsymbol{\Sigma})$ (Wishart), independent
- Inverse Wishart moment, finite for $T > N + 2$: $\mathbb{E}[\hat{\boldsymbol{\Sigma}}^{-1}] = \frac{T}{T-N-2}\boldsymbol{\Sigma}^{-1}$, an **upward** bias that grows with N/T
- Hence, with $\hat{\theta}^2 = \hat{\boldsymbol{\mu}}^\top \hat{\boldsymbol{\Sigma}}^{-1} \hat{\boldsymbol{\mu}}$ (Kan & Zhou (2007))

$$\mathbb{E}[\hat{\theta}^2] = \frac{T}{T-N-2} \left(\theta^2 + \frac{N}{T} \right) = \frac{T\theta^2 + N}{T-N-2}$$

- Sectors: $N = 9$, $T = 60$, true $\theta = 0.66$ a year: $\mathbb{E}[\hat{\theta}^2]$ corresponds to a Sharpe of 1.65 a year
 - The analytical counterpart of the simulated median of 1.56 on the previous slide

Correcting the Bias and the Distribution of Weights

- Unbiased estimator of θ^2 (Kan & Zhou (2007)): $\hat{\theta}_u^2 = [(T - N - 2)\hat{\theta}^2 - N]/T$
 - It can be negative (then truncated at zero), and $\sqrt{\hat{\theta}_u^2}$ is not unbiased for θ
- Sectors, full sample ($T = 331$, $N = 9$): plug-in $\hat{\theta} = 0.66$, bias-corrected $\sqrt{\hat{\theta}_u^2} = 0.31$ a year
 - The correction removes more than half of the in-sample tangency Sharpe, even with 27 years of data
- GMV weights: $\hat{w}_{\text{GMV}}$ is unbiased and its exact distribution and covariance are known (Okhrin & Schmid (2006))
- Tangency weights divide by $1^\top \hat{\Sigma}^{-1} \hat{\mu}$, which can be near zero: very heavy tails, the explosions of the rolling weights
- Estimation-risk-optimal mixtures of tangency, GMV and 1/N: Kan & Zhou (2007), Kan, Wang & Zhou (2022)

Tangency Weights by Regression: Britten-Jones (1999)

- Britten-Jones (1999): regress the constant 1 on the excess returns, **without** a constant term

$$1 = \mathbf{b}^\top \mathbf{r}_t + u_t, \quad \hat{\mathbf{b}} = (\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top \mathbf{1} \propto \hat{\Sigma}^{-1} \hat{\boldsymbol{\mu}}$$

- Sherman–Morrison on $\mathbf{X}^\top \mathbf{X} / T = \hat{\Sigma} + \hat{\boldsymbol{\mu}} \hat{\boldsymbol{\mu}}^\top$ gives the proportionality (Seminar 4, A2); $\hat{\mathbf{b}} / \mathbf{1}^\top \hat{\mathbf{b}} =$ sample tangency weights
- OLS t -test of $b_i = 0$: asset i has zero weight in the tangency portfolio
- F -test of $\mathbf{R}\mathbf{b} = 0$: for example $b_1 = \dots = b_N$, i.e. “the tangency portfolio is $1/N$ ”
- Exact finite-sample t and F distributions under i.i.d. Normal returns; HAC standard errors otherwise

Britten-Jones on the Nine Sectors

- Monthly excess returns, January 1999 – July 2026 ($T = 331$); normalised $\hat{b}$ reproduces the tangency weights (-0.37 financials, $+0.37$ health care)
- No single weight differs from zero: largest $|t| = 1.12$ (financials); with HAC errors 1.19
- H_0 : tangency = $1/N$, eight restrictions $b_i = b_9$
 - ▶ $F(8, 322) = 0.48$, $p = 0.87$; HAC Wald $\chi^2_8 = 4.50$, $p = 0.81$
- 27 years of data cannot tell the estimated optimum from equal weights
- The same verdict as the window-length simulation and the out-of-sample tests below

Unstable Weights



- Nine sector ETFs; each month the weights use only the previous 60 months (top panel clipped at ± 6)

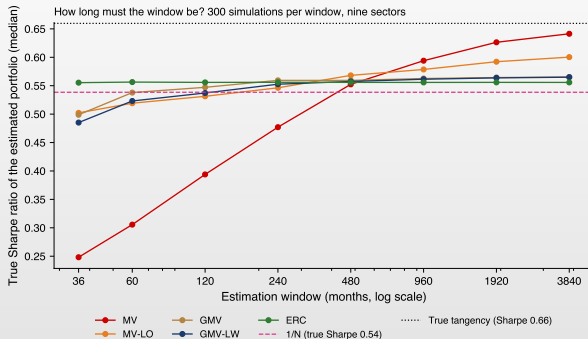
Reading the Rolling Weights

- ▣ MV: median gross exposure $\sum_i |w_i| = 4.9$; largest single weight 26 in absolute value
- ▣ Gross exposure above 3 in 90% of the months
- ▣ Explosions in 2004–2006 and 2009: $1^\top \hat{\Sigma}^{-1} \hat{\mu}$ close to zero in the denominator
 - ▶ Negative in 12 of 271 windows: MV, as published, then holds the minimum-Sharpe portfolio
- ▣ GMV with shrinkage (next section): gross exposure 1.5, largest weight 0.94
- ▣ Unstable weights mean high turnover, hence high trading costs

The 1/N Benchmark

- ▣ **1/N**: equal weights $w_i = 1/N$, rebalanced each month; no estimation at all
- ▣ DeMiguel, Garlappi & Uppal (2009): 14 optimisation models on 7 data sets
 - ▶ None beats 1/N consistently in Sharpe ratio, certainty-equivalent return or turnover
- ▣ Their calibration: about 3000 months of data for sample MV to beat 1/N with 25 assets
- ▣ Kan & Zhou (2007): combining tangency and GMV reduces estimation risk
- ▣ Question: how long must the window be for our nine sectors?

How Long Must the Window Be?



- 300 simulated samples per window length; true parameters as in the previous simulation

Reading the Window-Length Experiment

- ▣ MV beats 1/N only with 480 months (40 years) of data: median true Sharpe 0.55 vs 0.54
- ▣ With 60 months, MV reaches 0.31, less than 60% of 1/N
- ▣ Long-only MV, GMV and GMV with shrinkage are close to 1/N from 60 months on
- ▣ ERC (Section 6) is almost insensitive to the window: 0.56 throughout
- ▣ Here 1/N is nearly efficient: the true tangency Sharpe is only 0.66

Recap: The Limits of Markowitz

- ▣ Means are estimated with large errors; the optimiser amplifies them
- ▣ In-sample Sharpe ratios of optimised portfolios are biased upwards: $\mathbb{E}[\hat{\theta}^2] = (T\theta^2 + N)/(T - N - 2)$
- ▣ Britten-Jones regression: tangency weights are OLS coefficients normalised to sum to one; on sectors they are not statistically distinguishable from $1/N$
- ▣ Sample MV weights are unstable and extreme
- ▣ $1/N$ is a hard benchmark: MV needs decades of data to beat it
- ▣ Remedies: better covariance estimates, better means, or rules that avoid means

Why the Sample Covariance Matrix Fails

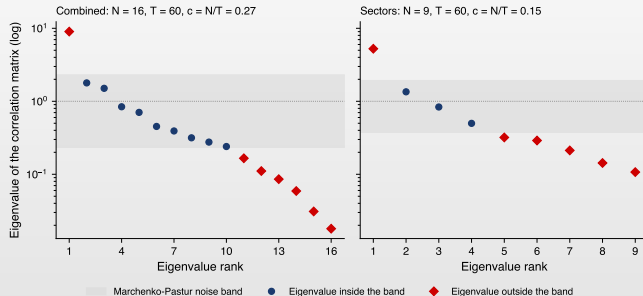
- ▣ N assets: $N(N + 1)/2$ variances and covariances; 16 assets: 136 parameters from 60 months
- ▣ Sample eigenvalues are too dispersed: the largest too large, the smallest too small
- ▣ **Condition number**: largest over smallest eigenvalue; a large value means Σ^{-1} amplifies noise
- ▣ Optimisers invert Σ , so they load on the directions with the smallest estimated variance
- ▣ The demeaned sample matrix has rank at most $T - 1$: singular, not invertible, whenever $T \leq N$

Eigenvalues of Pure Noise: Marchenko–Pastur

- i.i.d. returns with $\Sigma = \sigma^2 I$: every true eigenvalue equals σ^2
- Marčenko & Pastur (1967): as $N, T \rightarrow \infty$ with $N/T \rightarrow c \in (0, 1)$, the sample eigenvalues fill
$$[\lambda_-, \lambda_+] = [\sigma^2(1 - \sqrt{c})^2, \sigma^2(1 + \sqrt{c})^2]$$
- $c = 16/60 = 0.27$: band $[0.23, 2.30]$ for a correlation matrix ($\sigma^2 = 1$), a ten-fold spread from noise alone
- Laloux et al. (1999): eigenvalues outside the band are candidates for structure beyond this noise model; those inside cannot be distinguished from it by this comparison alone
- This turns “sample eigenvalues are too dispersed” into a quantitative benchmark

Sample Eigenvalues versus the Noise Band

Last 60-month window, Aug 2021 - Jul 2026



- Correlation matrices of monthly excess returns, last 60-month window (August 2021 – July 2026); grey band: $[\lambda_-, \lambda_+]$ for $\sigma^2 = 1$

Reading the Eigenvalue Chart

- ▣ 16 ETFs: one eigenvalue above $\lambda_+ = 2.30$, the market mode (9.0, 56% of the trace); nine lie inside the band
- ▣ Nine sectors: again only the market mode (5.2) lies above $\lambda_+ = 1.92$
- ▣ Rescaling the noise variance to $\sigma^2 = 1 - \lambda_1/N$ (Laloux et al. (1999)): three eigenvalues above the band in each universe
- ▣ Eigenvalues **below** the band (six of 16): near-riskless combinations, e.g. TLT against IEF
 - ▶ GMV loads exactly on these directions: if they are underestimated, its risk is too

The Shrinkage Estimator

- Shrink the sample matrix S towards a structured target F

$$\hat{\Sigma}_{\text{LW}} = \delta F + (1 - \delta) S, \quad 0 \leq \delta \leq 1$$

- Targets

- ▶ Single-index (market) model: [Ledoit & Wolf \(2003\)](#)
- ▶ Scaled identity: [Ledoit & Wolf \(2004a\)](#)
- ▶ Constant correlation, $f_{ii} = s_{ii}$ and $f_{ij} = \bar{r} \sqrt{s_{ii} s_{jj}}$ for $i \neq j$, $\bar{r}$ the average sample correlation: [Ledoit & Wolf \(2004b\)](#), used here

- δ trades the variance of S against the bias of F

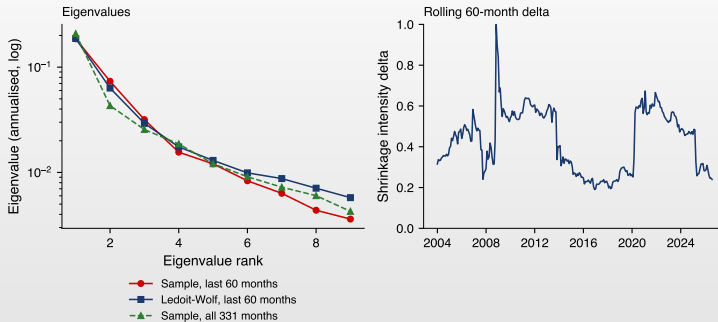
The Optimal Shrinkage Intensity

- Ledoit–Wolf choose δ to minimise the expected squared (Frobenius) distance to the true Σ

$$\delta^* = \max \left\{ 0, \min \left\{ 1, \frac{\hat{\kappa}}{T} \right\} \right\}, \quad \hat{\kappa} = \frac{\hat{\pi} - \hat{\rho}}{\hat{\gamma}}$$

- $\hat{\pi}$: sum of the asymptotic variances of the entries of S (noise)
- $\hat{\rho}$: their covariances with the target; $\hat{\gamma} = \|F - S\|^2$ (misspecification)
- Noisy data or a good target: large δ ; long samples: $\delta \rightarrow 0$

Shrinkage on US Sectors



- Nine sector ETFs, monthly excess returns; last window August 2021 – July 2026

Reading the Shrinkage Chart

- ▣ Last window: $\delta = 0.24$; the smallest eigenvalues are pulled up, the largest almost unchanged
- ▣ Condition number: sample 51.6, shrunk 32.5
- ▣ Rolling δ : mean 0.43, range 0.19–1.00
 - ▶ $\delta = 1$ in the early windows: 60 months are too few to beat the target
- ▣ GMV-LW: the GMV portfolio computed with $\hat{\Sigma}_{LW}$

Constraints Are Shrinkage Too

- ▣ Jagannathan & Ma (2003): the long-only GMV equals the unconstrained GMV of a modified covariance matrix

$$\tilde{S} = S - (\lambda \mathbf{1}^\top + \mathbf{1} \lambda^\top), \quad \lambda \geq 0 \text{ the multipliers of } w_i \geq 0$$

- ▶ $\tilde{s}_{ij} = s_{ij} - \lambda_i - \lambda_j$: a binding constraint on asset i lowers all its covariances, which shrinks the large estimates
- ▣ Imposing the “wrong” constraint (no short sales) can reduce out-of-sample risk
- ▣ The effect is largest when covariances are estimated from few observations
- ▣ Practical rule: shrink, constrain, or both, but never invert a raw sample matrix of many assets

Beyond Linear Shrinkage

- Linear shrinkage moves all eigenvalues towards the target with one δ ; **nonlinear** shrinkage corrects each eigenvalue separately
 - ▶ Ledoit & Wolf (2017): tuned for portfolio selection; Ledoit & Wolf (2020): an analytical formula, no optimisation
- Factor structure plus a sparse residual covariance: POET (Fan, Liao & Mincheva (2013)), the link to Chapter 3
- High-dimensional GMV with shrinkage towards a target portfolio: Bodnar, Parolya & Schmid (2018)
- Out of sample, 60-month window, GMV volatility
 - ▶ Sectors: sample 11.90%, linear (constant correlation) 11.80%, nonlinear 11.62%
 - ▶ 16 ETFs: sample 4.06%, linear 4.38%, nonlinear 4.31%: better than the linear target, not than the raw matrix

Recap: Covariance Shrinkage

- The sample covariance matrix is noisy and ill-conditioned when N is large relative to T
- Shrinkage mixes it with a structured target; the data choose the weight δ
- The target matters: constant correlation suits one asset class better than a mix of stocks and bonds
- No-short constraints act as implicit shrinkage
- Marchenko–Pastur gives an asymptotic noise benchmark for the eigenvalues; nonlinear shrinkage treats each eigenvalue separately

Shrinking the Means: One Family of Estimators

- Bayes–Stein (Jorion (1986)): $\hat{\mu}_{BS} = (1 - \hat{\phi}) \hat{\mu} + \hat{\phi} \hat{\mu}_{GMV}$
 - ▶ Empirical-Bayes shrinkage of every mean towards the mean of the GMV portfolio; $\hat{\phi}$ from a Stein argument, larger when N/T is large
- Tu & Zhou (2011): the convex combination of a sophisticated rule and $1/N$ that minimises the expected out-of-sample loss
- DeMiguel, Garlappi, Nogales & Uppal (2009): norm constraints $\|w\|_1 \leq \delta$ or $\|w\|_2 \leq \delta$ nest the long-only GMV, shrinkage GMV and $1/N$; the bridge to the lasso portfolios of Chapter 13
- Black–Litterman (next slides): Bayesian shrinkage of μ towards the returns implied by a benchmark

Black and Litterman: Starting from Equilibrium



Fischer Black (1938–1995)

Photo: Dalmatine at English Wikipedia; public domain; Wikimedia Commons

- Black & Litterman (1992), developed at Goldman Sachs
- Problem: sample means give extreme, unintuitive weights
- Idea: start from the returns that make a **benchmark** portfolio optimal
- Add the investor's **views**, weighted by their confidence
- Result: weights that move away from the benchmark only where views exist

Step 1: Reverse Optimisation

- ▣ A mean–variance investor with risk aversion δ_{RA} and access to cash holds $w^* = (\delta_{RA} \Sigma)^{-1} \mu$ in risky assets and $1 - \mathbf{1}^\top w^*$ in cash
- ▣ Reverse it: the returns that make the benchmark w_b optimal

$$\pi = \delta_{RA} \Sigma w_b$$

- ▣ π : **implied** (equilibrium) excess returns; with market weights, the CAPM returns
- ▣ Here: no market capitalisations for the sectors, so $w_b = 1/N$, $\delta_{RA} = 2.5$

Step 2: Views and the Posterior

- K views: $P\mu = q + \varepsilon$, $\varepsilon \sim N(0, \Omega)$; prior $\mu \sim N(\pi, \tau\Sigma)$
- Posterior mean: a precision-weighted average

$$\mu_{BL} = [(\tau\Sigma)^{-1} + P^\top \Omega^{-1} P]^{-1} [(\tau\Sigma)^{-1} \pi + P^\top \Omega^{-1} q]$$

- Weights: $w_{BL} = (\delta_{RA}\Sigma)^{-1} \mu_{BL}$
- He & Litterman (2002): $\tau = 0.05$, $\Omega = \text{diag}(P \tau \Sigma P^\top)$; with Σ in the weights, only the assets in the views move (the posterior covariance would also scale all weights by $1/(1 + \tau)$)

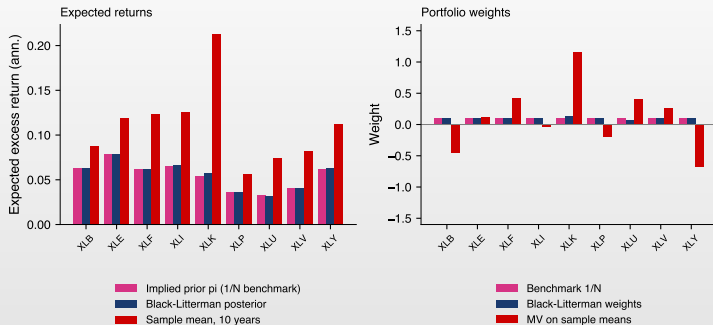
Black–Litterman as Mixed (GLS) Estimation

- Theil & Goldberger (1961): stack the prior and the views as one regression on μ

$$\begin{pmatrix} \pi \\ q \end{pmatrix} = \begin{pmatrix} I \\ P \end{pmatrix} \mu + \begin{pmatrix} e_1 \\ e_2 \end{pmatrix}, \quad \text{Var} \begin{pmatrix} e_1 \\ e_2 \end{pmatrix} = \begin{pmatrix} \tau \Sigma & 0 \\ 0 & \Omega \end{pmatrix}$$

- GLS gives exactly μ_{BL} of the previous slide, with covariance $M = [(\tau \Sigma)^{-1} + P^T \Omega^{-1} P]^{-1}$
- Equivalent form: $\mu_{BL} = \pi + \tau \Sigma P^T (P \tau \Sigma P^T + \Omega)^{-1} (q - P \pi)$
- Limits: $\Omega \rightarrow 0$ gives $P \mu_{BL} = q$ exactly (restricted least squares); $\Omega \rightarrow \infty$ gives π
- One asset, $\pi = 5\%$, $\tau \sigma^2 = 0.00128$, view 8% with variance 0.0004: $\mu_{BL} = 7.29\%$, view weight 0.76

Black-Litterman on US Sectors



□ Σ from monthly excess returns, August 2016 – July 2026; one view: XLK minus XLU = 3% a year

Reading Black–Litterman

- ▣ Implied XLK minus XLU: 2.2%; view 3%; posterior 2.6% (the average, since Ω equals the prior variance of the view)
- ▣ Weights: XLK from 11.1% to 14.4%, XLU from 11.1% to 7.8%, all others unchanged
- ▣ Sample means instead: XLK 21.3% a year; MV weights +115% XLK, −68% XLY, −46% XLB
- ▣ The view is relative ($P1 = 0$): the tilt $\propto P^\top$ sums to zero, so the portfolio stays fully invested; an absolute view would also move the cash position
- ▣ Black–Litterman is Bayesian shrinkage of the means towards equilibrium

Recap: Black–Litterman

- The prior comes from a benchmark, not from noisy sample means
- Views enter with explicit confidence; weights move only where views exist
- Results depend on the benchmark, τ , Ω and the risk aversion
- Without views, Black–Litterman returns the benchmark itself
- Formally a GLS (mixed) estimator: one member of the family of mean-shrinkage estimators with Bayes–Stein and 1/N combinations

Risk Contributions

- ▣ Volatility is homogeneous of degree one in the weights (Euler)

$$\sigma_p = \sum_{i=1}^N w_i \frac{\partial \sigma_p}{\partial w_i} = \sum_{i=1}^N \frac{w_i (\Sigma w)_i}{\sigma_p}$$

- ▣ **Risk contribution** of asset i : $RC_i = w_i (\Sigma w)_i / \sigma_p$; percentage share RC_i / σ_p
- ▣ Equal weights are not equal risks: volatile assets dominate
- ▣ Risk budgeting: choose w so that the shares RC_i / σ_p equal preset budgets b_i

Equal Risk Contribution

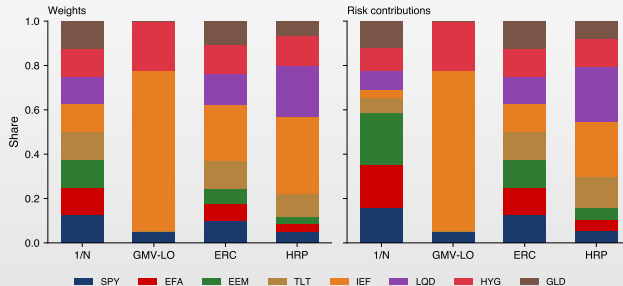
- ERC (Maillard, Roncalli & Teïletche (2010)): $RC_i = RC_j$ for all i, j
- Computed from a convex problem, then normalised to sum to one

$$\min_{y>0} \frac{1}{2} y^\top \Sigma y - \frac{1}{N} \sum_i \ln y_i, \quad w = y / 1^\top y$$

- Volatility ordering: $\sigma_{\text{GMV}} \leq \sigma_{\text{ERC}} \leq \sigma_{1/N}$
- Two assets: ERC is **inverse volatility**, $w_1 = \sigma_1^{-1} / (\sigma_1^{-1} + \sigma_2^{-1})$, for any $\rho \in (-1, 1)$
 - SPY (14.7%) and IEF (6.6%): ERC holds 31% SPY; a 60/40 mix lets SPY carry 94% of the risk
- Needs no expected returns, only Σ ; for Σ positive definite a solution exists and is unique (strictly convex objective, infinite at the boundary)

Weights versus Risk Contributions

Eight multi-asset ETFs, monthly May 2007 - Jul 2026



- Eight multi-asset ETFs, full-sample covariance matrix, May 2007 – July 2026 ($T = 231$)

Reading the Risk Contributions

- ▣ 1/N: equities and high yield receive half of the weight but 69% of the risk
- ▣ ERC: 12.5% of the risk from every asset; IEF gets the largest weight (25%)
- ▣ Long-only GMV: 73% IEF, 22% HYG, 5% SPY; five assets at zero
- ▣ Volatility: 1/N 9.4%, ERC 7.9%, HRP 7.1%, long-only GMV 5.8%
- ▣ ERC lies between 1/N and GMV, as the theory predicts

Hierarchical Clustering of Assets

- ▣ Correlation distance: $d_{ij} = \sqrt{(1 - \rho_{ij})/2} \in [0, 1]$
- ▣ **Single linkage**: repeatedly merge the two closest clusters; cluster distance = closest pair
- ▣ The result is a tree (**dendrogram**); cutting it gives groups of similar assets
- ▣ Mantegna (1999): the same distance and the **minimum spanning tree** reveal the hierarchy of a stock market
- ▣ Textbook: Härdle & Simar (2019), cluster analysis

Hierarchical Risk Parity

▣ HRP (López de Prado (2016)), three steps

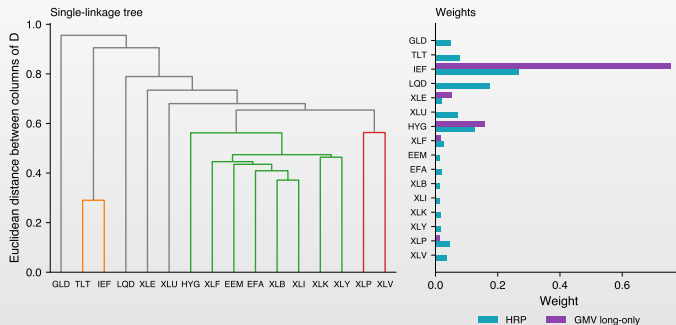
- ▶ Tree clustering: Euclidean distance between the columns of $D = (d_{ij})$, $\tilde{d}_{ij} = \sqrt{\sum_k (d_{ki} - d_{kj})^2}$, then single linkage on $\tilde{d}_{ij}$
- ▶ Quasi-diagonalisation: reorder assets so that similar ones are adjacent
- ▶ Recursive bisection: split the ordered list in two; split the weight in inverse proportion to the variance of each half

▣ Variance of a half: inverse-variance weights inside it, $\tilde{w}^\top \Sigma \tilde{w}$

▣ No matrix inversion: works even when Σ is singular

▣ All weights are positive

HRP on 16 ETFs



- Nine sectors and seven multi-asset ETFs, monthly, May 2007 – July 2026

Reading the Tree

- ▣ Closest pair: TLT and IEF ($\tilde{d} = 0.29$); gold joins last (0.96)
- ▣ Equity cluster: cyclical sectors, EFA, EEM and high-yield credit
- ▣ Defensive sectors (utilities, staples, health care) and energy join the equity cluster late
- ▣ HRP: all 16 weights positive, largest IEF 26.7%, LQD 17.6% and HYG 12.6%
- ▣ Long-only GMV: 75.6% in IEF and only five assets above zero

Other Risk-Based Rules

- ▣ **Inverse volatility:** $w_i \propto 1/\sigma_i$; equals ERC when all correlations are equal
- ▣ **Maximum diversification** (Choueifaty & Coignard (2008)): maximise $w^\top \sigma / \sqrt{w^\top \Sigma w}$
- ▣ **Minimum variance:** GMV, the extreme risk-based rule
- ▣ All ignore expected returns: robust to estimation error, but optimal only under special assumptions
 - ▶ ERC is the tangency portfolio if all assets have the same Sharpe ratio and equal correlations

Recap: Risk-Based Allocation

- ▣ Risk contributions decompose portfolio volatility additively
- ▣ ERC equalises them; it lies between $1/N$ and GMV
- ▣ HRP uses a correlation tree and needs no matrix inversion
- ▣ These rules avoid expected returns entirely: less estimation error, no view on returns

Backtest Design

- ▣ **Out-of-sample** (OOS): weights at month t use only data up to $t - 1$
- ▣ Rolling window of 60 months; rebalancing every month
- ▣ Eight rules: 1/N, MV, MV-LO, GMV, GMV-LO, GMV-LW, ERC, HRP
- ▣ Three universes: sectors (OOS January 2004 – July 2026, 271 months), multi-asset and combined (May 2012 – July 2026, 171 months)
- ▣ Annualised mean, volatility and Sharpe ratio of excess returns; maximum drawdown of total-return wealth

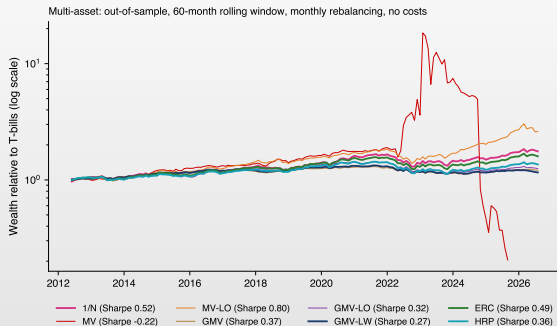
Turnover and Transaction Costs

- Between rebalancings weights drift with returns

$$w_{i,t}^+ = \frac{w_{i,t}(1 + R_{i,t})}{\sum_j w_{j,t}(1 + R_{j,t})}$$

- Turnover:** traded fraction of wealth, $TO_{t+1} = \sum_i |w_{i,t+1} - w_{i,t}^+|$
- Net return with cost c per unit traded: $R_{p,t}^{\text{net}} = R_{p,t} - c TO_t$
- 1/N also trades: it sells winners and buys losers every month

Out-of-Sample Growth: Multi-Asset

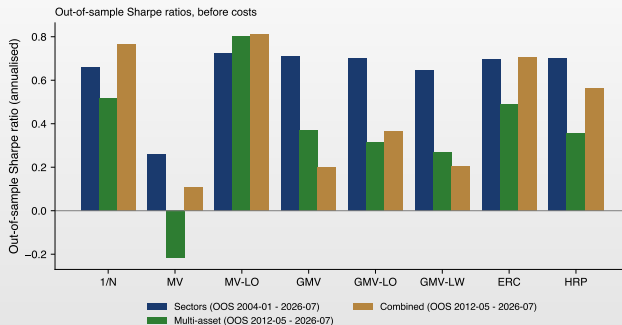


- Eight multi-asset ETFs; wealth relative to a Treasury-bill account, before costs; a path ends at a monthly loss of 100% or more (MV)

Reading the Multi-Asset Backtest

- ▣ MV collapses: annualised volatility 518%, turnover 44 times wealth per month; a monthly loss above 100% ends its wealth path
- ▣ MV-LO has the highest Sharpe (0.80); 1/N 0.52; ERC 0.49
- ▣ GMV rules have the lowest volatility (3.8%–5.4%) but also low Sharpe ratios (0.27–0.37)
 - ▶ They load on Treasuries, whose excess returns were slightly negative after May 2012 (TLT -0.5% , IEF -0.2% a year)
- ▣ Is MV-LO really better, or lucky? A test is needed

Sharpe Ratios Across Universes



60-month rolling window, monthly rebalancing, annualised, before costs

MFM_ch4_oos_backtest

Results: Sector Universe

Rule	Mean	Volatility	Sharpe	Turnover	Max. drawdown
1/N	9.4%	14.3%	0.66	0.02	−49%
MV	17.8%	69.0%	0.26	3.86	−100%
MV-LO	10.7%	14.7%	0.72	0.23	−42%
GMV	8.5%	11.9%	0.71	0.23	−33%
GMV-LO	8.1%	11.5%	0.70	0.09	−34%
GMV-LW	7.6%	11.8%	0.65	0.14	−31%
ERC	9.3%	13.2%	0.70	0.03	−47%
HRP	8.9%	12.8%	0.70	0.08	−45%

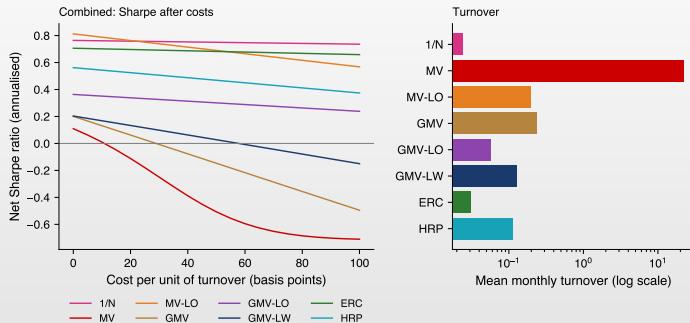
- January 2004 – July 2026; turnover: mean monthly traded fraction of wealth; drawdown of total-return wealth
- MV: a monthly loss above 100% wipes out the capital (drawdown −100%); its mean and Sharpe describe a hypothetical return series, not investable wealth
- All rules except MV land within 0.07 of 1/N

Results: Combined Universe (16 ETFs)

Rule	Mean	Volatility	Sharpe	Turnover	Sharpe, 50 bp
1/N	7.9%	10.3%	0.76	0.02	0.75
MV	28.7%	261.8%	0.11	22.10	-0.51
MV-LO	8.0%	9.8%	0.81	0.20	0.69
GMV	0.8%	4.1%	0.20	0.24	-0.15
GMV-LO	2.0%	5.4%	0.36	0.06	0.30
GMV-LW	0.9%	4.4%	0.20	0.13	0.03
ERC	5.5%	7.8%	0.71	0.03	0.68
HRP	4.0%	7.0%	0.56	0.11	0.47

- May 2012 – July 2026; last column: net of 50 basis points per unit of turnover
- 1 basis point (bp) = 0.01%

Sharpe Ratios after Costs



Combined universe (16 ETFs); cost c from 0 to 100 bp per unit of turnover

Reading the Cost Chart

- ▣ Turnover ranges from 0.02 (1/N) to 22 (MV) per month
- ▣ Net Sharpe turns negative at about 11 bp for MV, 29 bp for GMV, 58 bp for GMV-LW
- ▣ MV-LO overtakes 1/N only for costs below about 22 bp (Seminar, B8)
- ▣ 1/N and ERC barely move: low turnover is a hidden advantage
- ▣ Rule: report net performance for realistic costs, not only gross

Inference on One Sharpe Ratio: Lo (2002)

- Lo (2002), i.i.d. Normal returns: $\sqrt{T}(\widehat{SR} - SR) \rightarrow N(0, 1 + SR^2/2)$, with SR monthly
- 1/N on sectors, $T = 271$: monthly $\widehat{SR} = 0.191$, standard error 0.061
 - ▶ Annualised $0.66 \pm 1.96 \times 0.21$: 95% interval $[0.25, 1.08]$
- Annualising by $\sqrt{q}$ ($q = 12$) is exact only without autocorrelation; in general $SR(q) = \eta(q) SR$

$$\eta(q) = \frac{q}{\sqrt{q + 2 \sum_{k=1}^{q-1} (q-k) \rho_k}}$$

- Here $\rho_1 = -0.03$, $\sum_{k=1}^{11} \rho_k = -0.11$: $\eta = 3.62$ instead of $\sqrt{12} = 3.46$, annual Sharpe 0.69

Testing a Difference in Sharpe Ratios

- H_0 : $SR_1 = SR_2$ for two strategies evaluated on the same months
- Jobson & Korkie (1981): a test under i.i.d. Normal returns
- Ledoit & Wolf (2008): a robust version
 - ▶ $\hat{\Delta} = f(\hat{v})$, $v = (\mu_1, \mu_2, \mathbb{E}[r_1^2], \mathbb{E}[r_2^2])$, $f = \frac{\mu_1}{\sqrt{\mathbb{E}r_1^2 - \mu_1^2}} - \frac{\mu_2}{\sqrt{\mathbb{E}r_2^2 - \mu_2^2}}$
 - ▶ Standard error by the delta method: $\sqrt{\nabla f^\top \hat{\Psi} \nabla f / T}$, $\hat{\Psi}$ a HAC covariance
- Strategies on the same assets are highly correlated: small differences can be significant

The Block Bootstrap



Bradley Efron, inventor of the bootstrap
(1979)

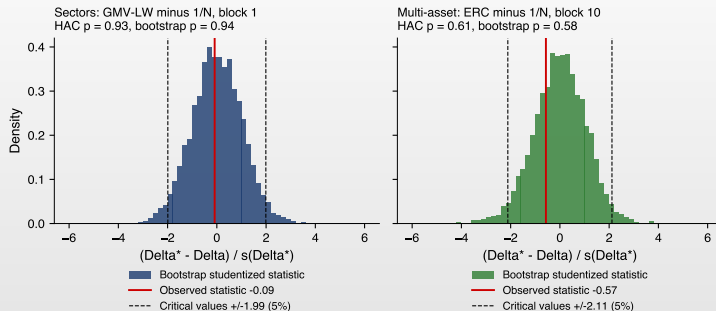
Photo: Ryan K. Morris (2007); public
domain; Wikimedia Commons

- Bootstrap: resample the data to approximate the sampling distribution of $\hat{\Delta}$
- Returns are dependent (volatility clustering): resample **blocks** of consecutive months (Künsch (1989))
- Ledoit & Wolf (2008), Section 3.2.2: circular blocks of pairs (r_{1t}, r_{2t}) , wrapping around the end of the sample
- **Studentized**: bootstrap $|\hat{\Delta}^* - \hat{\Delta}|/s(\hat{\Delta}^*)$, with $s(\hat{\Delta}^*)$ from the block sums of each resample
 - ▶ The studentized statistic is asymptotically pivotal: more accurate than the Normal approximation of the HAC test
- $s(\hat{\Delta})$ on the data: prewhitened quadratic-spectral kernel (Andrews & Monahan (1992)), automatic bandwidth (Andrews (1991))
- $p = [\#\{\tilde{d}^{*m} \geq |\hat{\Delta}|/s(\hat{\Delta})\} + 1]/(M + 1)$, $M = 4999$ draws

Choosing the Block Length: Algorithm 3.1

- Ledoit & Wolf (2008), Algorithm 3.1: choose the block b whose bootstrap interval has actual coverage closest to 95%
 - ▶ Fit a first-order vector autoregression to the pair of returns; resample its residuals by a stationary bootstrap (Politis & Romano (1994)), mean block 5
 - ▶ Generate $K = 5000$ pseudo-samples of length T ; in each, build the 95% interval for $b \in \{1, 2, 4, 6, 8, 10\}$
 - ▶ $\hat{g}(b)$: share of intervals that contain the pseudo-true $\hat{\Delta}$; pick $\tilde{b} = \arg \min_b |\hat{g}(b) - 0.95|$
- The grid and K are those of the paper (Sections 4–5); applied to each of the 21 pairs of this chapter
- Chosen blocks: mostly 1 for sectors, 8–10 for multi-asset and combined rules (stronger dependence)
- The earlier non-studentized bootstrap with a fixed 6-month block is not the published test and is no longer used

Is Anything Better than $1/N$?



- Bootstrap distributions of the studentized Sharpe ratio difference against $1/N$; red: observed statistic; dashed: bootstrap critical values

Reading the Tests

- ▣ Sectors, GMV-LW minus $1/N$: -0.02 , 95% interval $[-0.36, 0.33]$; HAC $p = 0.93$, bootstrap $p = 0.94$ (block 1)
- ▣ Multi-asset, ERC minus $1/N$: -0.03 , interval $[-0.15, 0.08]$; HAC $p = 0.61$, bootstrap $p = 0.58$ (block 10)
 - ▶ ERC and $1/N$ are so correlated that the interval is narrow, yet it still covers zero
- ▣ Sectors: no rule differs from $1/N$ at 5%; the closest is ERC, $+0.04$ (HAC $p = 0.09$, bootstrap 0.13)
- ▣ Multi-asset: MV-LO $+0.29$ is **not** significant (HAC $p = 0.12$, bootstrap 0.16)

Significant Differences: Mostly Losses

Universe	Rule minus 1/N	$\hat{\Delta}$	HAC p	Bootstrap p
Multi-asset	MV	-0.74	0.01	0.06
Multi-asset	HRP	-0.16	0.07	0.05
Combined	GMV	-0.57	0.03	0.05
Combined	GMV-LO	-0.40	0.05	0.08
Combined	GMV-LW	-0.56	0.02	0.03
Combined	MV-LO	+0.05	0.82	0.82

- No rule beats 1/N significantly in any universe
- GMV rules lose significantly on the combined universe: low risk, but too little return
- 21 tests: Holm-adjusted p -values are at least 0.29 (HAC) and 0.68 (bootstrap): nothing survives (Chapter 3, multiple testing)

Many Rules, One Winner: the Deflated Sharpe Ratio

- ▣ Bailey & López de Prado (2014): the best of N Sharpe ratios is biased upwards even if every true Sharpe ratio is zero
- ▣ Expected maximum under H_0 : all true Sharpe ratios zero, N independent trials, monthly units as T ; with $\gamma_E = 0.5772$ (Euler's constant) and $\hat{V}$ the variance of the N estimates

$$SR_0 = \sqrt{\hat{V}} \left[(1 - \gamma_E) \Phi^{-1} \left(1 - \frac{1}{N} \right) + \gamma_E \Phi^{-1} \left(1 - \frac{1}{Ne} \right) \right]$$

- ▣ Deflated Sharpe ratio, with skewness $\hat{\gamma}_3$ and kurtosis $\hat{\gamma}_4$ of the winner

$$DSR = \Phi \left(\frac{(\widehat{SR} - SR_0) \sqrt{T-1}}{\sqrt{1 - \hat{\gamma}_3 \widehat{SR} + \frac{\hat{\gamma}_4 - 1}{4} \widehat{SR}^2}} \right)$$

- ▣ This chapter: 24 configurations (8 rules $\times$ 3 universes); winner MV-LO on 16 ETFs, 0.81 a year
 - ▶ As 24 independent trials: $SR_0 = 0.52$ a year, $DSR = 0.84 < 0.95$
 - ▶ The configurations are strongly correlated (up to 0.99), so fewer are independent: $N = 8$ gives DSR 0.93, $N = 3$ gives 0.98; the verdict hinges on the effective N

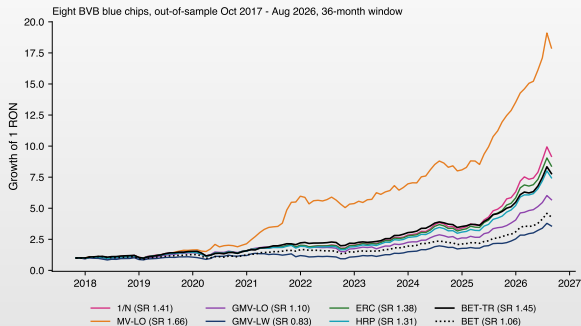
Recap: Out-of-Sample Evaluation

- ▣ Only out-of-sample, net-of-cost results are informative
- ▣ Sample MV fails in every universe; constraints and shrinkage tame it
- ▣ Risk-based rules and $1/N$ are hard to separate statistically
- ▣ Test Sharpe differences with HAC errors or the studentized block bootstrap, and count your tests
- ▣ Single Sharpe ratios need standard errors and autocorrelation-aware annualisation; a winner among many needs deflation

Setup: a Romanian Investor

- ▣ Eight blue chips listed since 2014: TLV, SNP, BRD, TGN, SNG, SNN, EL, TEL
- ▣ Monthly returns in RON, October 2014 – August 2026; 36-month window, first portfolio October 2017
- ▣ Rules: 1/N, MV-LO, GMV-LO, GMV-LW, ERC, HRP; benchmarks BET-TR and BET
- ▣ Sharpe ratios with $R_f = 0$ (no RON money-market series in the data): return per unit of risk, higher than excess-return Sharpe ratios
- ▣ Earlier Romanian evidence on diversification: Dragotă, Șerbănescu & Pele (2008)

BVB Portfolios versus BET-TR



□ Out-of-sample, October 2017 – August 2026 (107 months), before costs

Reading the BVB Backtest

- ▣ BET-TR: mean 24.7% a year, volatility 17.0%, Sharpe 1.45; BET: mean 17.8%, Sharpe 1.07
 - ▶ The gap of about 7% a year is the dividends: compare with BET-TR, never with BET
- ▣ 1/N 1.41, ERC 1.38, HRP 1.31: close to BET-TR
- ▣ MV-LO 1.66 (difference +0.21, $p = 0.44$); GMV-LW 0.83 (-0.63 , $p = 0.05$)
- ▣ Average correlation of the eight stocks: 0.44; the latest GMV-LO puts 70% in TLV

Beyond Equities: Digital Assets

- ▣ [Petukhina, Trimborn, Härdle & Elendner \(2021\)](#): cryptocurrencies in portfolio allocation rules; gains depend on the rule and on liquidity
- ▣ [Pele et al. \(2023\)](#): are cryptos becoming alternative assets?
- ▣ Bitcoin: annualised volatility 172% (monthly, 2010–2026), IEF 6.6%
- ▣ Risk contributions grow with volatility: risk-based rules give Bitcoin a small weight automatically
- ▣ Robust mean–variance for crypto-assets: [Pele \(2023\)](#); Seminar 4, B5

Recap: BVB Portfolios

- Benchmark choice matters: BET omits about 7% a year of dividends
- No optimised BVB-only rule beats BET-TR significantly
- Eight stocks with average correlation 0.44: little room for diversification inside the BVB
- Open question: does diversification **beyond** BET pay? (Seminar 4, Part C)

Case Study: Which Frontier Can a Large Investor Reach?

- Jensen, Kelly, Malamud & Pedersen (2026), *Review of Financial Studies*: how should machine-learning return forecasts become portfolios when every trade moves the price?
- The rules of this chapter fight estimation risk; Ledoit & Wolf (2017) is the landmark for the risk input $\hat{\Sigma}$, but costs enter at most as a flat rate per unit of turnover
- Implementable efficient frontier (eq. 12): for each volatility σ , the highest expected return **net of trading costs**
 - ▶ $k(\sigma) = \max_{\pi} E[r_{t+1}^{\text{net}}]$ subject to $E[\pi_t^{\top} \Sigma \pi_t] = \sigma^2$
 - ▶ Proposition 1: the net Sharpe ratio $k(\sigma)/\sigma$ falls along the frontier, and a larger investor faces a lower frontier
- Usual two-step practice: forecast returns with ML, then optimise; the forecasts load on short-lived signals of small stocks, whose turnover eats the return
- The answer of the paper generalises the dynamic trading rule of Gârleanu & Pedersen (2013) and learns the weights directly from the signals, as Brandt, Santa-Clara & Valkanov (2009) do without costs

Case Study: Data and Method as Published

- Trading cost as a fraction of wealth W_t (eqs. 4, 7): $c_t = \frac{W_t}{2} (\pi_t - g_t \pi_{t-1})^\top \Lambda (\pi_t - g_t \pi_{t-1})$
 - ▶ π_t : weights; g_t : drift of last month's weights with the returns; $\Lambda = \text{diag}(0.2/V_i)$, V_i the average daily dollar volume over 6 months: trading 1% of V_i moves the price by 0.1% (eq. 35, Frazzini, Israel & Moskowitz (2018))
- Optimal rule (Proposition 3): $\pi_t = m g_t \pi_{t-1} + (1 - m) A_t$: trade part of the way towards an aim portfolio A_t , a weighted sum of today's and expected future Markowitz portfolios; m solves eq. (14)
- Portfolio-ML (Proposition 4, eq. 40): $A_t = \text{diag}(1/\sigma_{i,t}) f(s_t) \beta$, $f = p$ random Fourier features of the signals s_t
 - ▶ $\hat{\beta} = (E_T[\tilde{\Sigma}_t] + \lambda I)^{-1} E_T[\tilde{r}_{t+1}]$: a ridge regression that maximises utility net of costs, in closed form (eqs. 24–26)
- Data (Section 5.1): NYSE common stocks above the median market capitalisation (184–805 stocks), 115 characteristics of Jensen, Kelly & Pedersen (2023) ranked into $[0, 1]$; December 1952 – December 2020, out of sample 1981–2020, monthly
- Σ : 13 characteristic themes and 12 industries as factors, exponential weights with half-lives of 378 days (correlations) and 126 days (variances); investor: $\gamma = 10$, wealth \$10 billion at the end of 2020, growing with the market
- Tuning (Table 1): $\lambda \in \{0, e^{-10}, e^{-9.8}, \dots, e^{10}\}$, $p \in \{2^6, \dots, 2^9\}$, $\eta \in \{e^{-3}, e^{-2}\}$ (standard deviation of the random weights), chosen each year by the utility of validation backtests since 1971
- Rivals: Markowitz-ML (eq. 32), Static-ML (one-period cost, eq. 34), Multiperiod-ML (forecasts for 12 horizons, eq. 22), factor sorts, $1/N$, minimum variance, market; “*” marks an extra tuning layer (Section 5.3)

Case Study: Key Results

- Table 2, Markowitz-ML: gross Sharpe ratio 2.06, but monthly turnover 79.36 and leverage 74.61: net return extremely negative
 - ▶ The textbook frontier exists only on paper
- Table 2, Portfolio-ML: net Sharpe ratio 1.33, utility 0.086 (net return 15%, volatility 11%, cost 0.6% a year, turnover 0.25)
 - ▶ Utility (eq. 11): annualised net return minus $\frac{\gamma}{2}$ times the annualised variance
- Table 2, rivals (net Sharpe ratio, utility): Static-ML* 0.81, 0.030; Multiperiod-ML* 0.83, 0.020; 1/N 0.54, -0.051; market 0.51, -0.033
 - ▶ Only cost-aware rules deliver positive utility; Portfolio-ML needs no second tuning layer
- Table 3: posterior probability that Portfolio-ML has the higher utility: 99.9% against Static-ML* and Multiperiod-ML*, 100% against Markowitz-ML
- Figure 1(B) (Section 6.2), $\gamma = 10$, volatility 11%: net excess return 14.6% with \$10 billion, 16.3% with \$1 billion
 - ▶ The frontier depends on who invests: size is a cost
- Figure 5: net of costs, quality is the most important theme and short-term reversal almost irrelevant; monthly autocorrelation -0.04 for 1-month reversal, 0.94 for book-to-market (Section 6.3)
 - ▶ Persistent signals are cheap to trade

Case Study: Replication, Limits and an Open Extension

- Course data: 14 US stocks and 9 sector ETFs (data/market); five signals from prices and volumes (1-, 6- and 12-month returns, 1-year volatility, 6-month dollar volume); same cost rule, γ , wealth and grid; Σ with the same weights on the 23 assets; validation from 2011, out of sample February 2014 – August 2026 (151 months)
- Portfolio-ML: net Sharpe ratio 0.57, utility 0.016, cost 0.6% a year, leverage 0.32; 1/N: net Sharpe ratio 0.61, but utility -0.26 (net volatility 30%)
 - ▶ Static-ML -0.13 ; minimum variance -0.73 (cost 43% a year); Markowitz-ML extremely negative (leverage 18)
 - ▶ Posterior probability of a higher utility for Portfolio-ML: 99.5% against Static-ML, 99.9% against minimum variance, 93.3% against 1/N
- Same ranking of cost-aware against cost-agnostic rules as in Table 2; Portfolio-ML wins with small, slowly traded positions, not with a higher gross Sharpe ratio (0.68 against 0.87 for 1/N)
- Limits of the course data: 23 assets instead of 184–805, five signals instead of 115, three validation years instead of ten, no factor model for Σ
- Limits stated by the authors: fees and other real-world frictions, computing power and ML methods not available in real time early in the sample (Section 6.1); costs and risk treated as constant over time (Section 5.1.3); among the smallest 20% of stocks Portfolio-ML ranks behind Multiperiod-ML* and Static-ML* (Figure 8)
- Open extension: implementable frontiers on the course data for wealth of \$1, 10 and 100 billion and $\gamma \in \{1, 5, 10, 20, 100\}$: where does 1/N fall relative to them? (Seminar 4, C3)

An Open Question

- Does any **estimated** allocation rule beat $1/N$ out of sample, net of costs, once every rule, window and universe tried is counted?
- Live evidence points both ways
 - ▶ Nonlinear shrinkage of Σ improves GMV out of sample in large cross-sections: [Ledoit & Wolf \(2017\)](#)
 - ▶ Combining tangency, GMV and $1/N$ under estimation risk: [Kan, Wang & Zhou \(2022\)](#)
 - ▶ Backtests need haircuts for the number of specifications tried: [Harvey & Liu \(2015\)](#)
- Formal statement for K rules on one universe
 - ▶ H_0 : $\Delta_k = SR_k - SR_{1/N} \leq 0$ for all k ; H_1 : $\Delta_k > 0$ for some k
 - ▶ Statistic: $\hat{\Delta}_k$ with HAC or block-bootstrap errors ([Ledoit & Wolf \(2008\)](#)); error rate fixed for the whole family
- Falsified by: a positive $\hat{\Delta}_k$ that survives a family-wise correction, costs and a second sample

The Discovery Loop, Step by Step (1/2)

- Tools are interchangeable: an AI assistant (Claude, ChatGPT, Gemini, GitHub Copilot) and literature engines (Semantic Scholar, Elicit, Google Scholar)
- **Literature:** *"List peer-reviewed studies since 2015 comparing shrinkage or risk-based rules with $1/N$ out of sample; for each give DOI, N , T , cost assumption and test."*
 - ▶ Resolve every DOI on Crossref and open the table cited; assistants merge authors and invent DOIs
- **Hypotheses:** *"Give three competing explanations for a Sharpe gain over $1/N$ (better Σ , a low-volatility premium, luck) and the testable implication of each for turnover, subperiods and factor loadings."*
- **Data and code:** *"Write Python that reads data/market, joins prices on common dates and runs a 60-month rolling GMV-LW and $1/N$; weights at t may use data up to $t - 1$ only."*
 - ▶ Replicate first: the combined-universe table of this chapter ($1/N$ 0.76, GMV-LW 0.20) before changing anything

The Discovery Loop, Step by Step (2/2)

- **Robustness:** *"Draft a pre-registration: the family of tests ($\text{rules} \times \text{universes} \times \text{windows} \times \text{costs}$), the error rate (FWER or FDR), the subperiods and a placebo, fixed before any result is seen."*
 - ▶ Placebo: random long-only weights with the same turnover; a real effect must beat their distribution
- **Critique:** *"Act as a referee: give the three strongest reasons my Sharpe gain is spurious and the test that would expose each."*
- **Report:** every specification tried, adjusted p -values, the power of the design, code and seeds
- Fix the plan before the first run: any specification chosen after seeing results is exploratory and is reported as such

What the Researcher Must Check

- **Look-ahead:** weights at t built with month- t returns $\Rightarrow$ assert that the last estimation date is before t ; lagging the weights one more month is only a sensitivity check
- **Survivorship and selection:** an ETF list chosen today $\Rightarrow$ use the universe available at each date, including later delisted ETFs and their final returns; otherwise disclose the limitation
- **Multiple testing:** 7 rules $\times$ 3 universes already give 21 tests $\Rightarrow$ Holm (FWER) or Benjamini–Hochberg (FDR, [Benjamini & Hochberg \(1995\)](#)) adjusted p -values
- **Power:** $\hat{\Delta}$ has a standard error near 0.18 on 171 months $\Rightarrow$ compute the detectable effect before running
- **Return type:** portfolios aggregate **simple** returns; Sharpe ratios use **excess** returns $\Rightarrow$ recompute one month step by step
- **Invented references:** an unresolvable DOI or a title mismatch $\Rightarrow$ drop the claim, log it in `AI_ERRORS.md`

Mini-Case: Data Snooping in This Chapter's Backtest

- A typical assistant proposal: “MV-LO beats 1/N on the multi-asset universe”; $H_0: \Delta \leq 0$, one-sided $p = 1 - \Phi(\hat{\Delta}/\text{se})$
- Replicate: $\hat{\Delta} = +0.29$, HAC standard error 0.18, one-sided $p = 0.06$ (two-sided HAC 0.12, bootstrap 0.16, Section 7)
- Test the family, not the winner: 21 tests (7 rules $\times$ 3 universes)
 - ▶ Only 7 of 21 differences are positive; the 4 two-sided HAC p -values below 0.05 are all **losses**, which never reject a superiority null
 - ▶ One-sided: smallest $p = 0.046$ (sectors, ERC); adjusted Holm 0.96, Benjamini–Hochberg 0.62; 0 discoveries at an FDR of 10%
- Power 0.8 for $\Delta = 0.29$, one-sided: about 435 months (36 years) at 5%, 944 months (79 years) at the Holm level 5%/21
- Interpretation: the data cannot rank these rules; the bootstrap interval for MV-LO, $[-0.13, 0.70]$, admits losses and large gains; failure to reject is not proof of equality

Your Turn: A Team-Project Seed

- Question: does nonlinear shrinkage ([Ledoit & Wolf \(2017\)](#)) or a three-fund combination ([Kan, Wang & Zhou \(2022\)](#)) beat $1/N$ on the 16 ETFs of this chapter, after 10–50 bp costs?
- Replicate first: the combined-universe table (1/N Sharpe 0.76, MV-LO 0.81) and the adjusted p -values of the mini-case
- Deliverables
 - ▶ A one-page pre-registration: hypotheses, family of tests, error rate, power, placebo
 - ▶ Code on data/market, a table of all specifications, a referee-style self-critique
- Log every prompt, tool and accepted output in `AI_USE.md`; every error caught (invented DOI, look-ahead, wrong return type) and the test that caught it in `AI_ERRORS.md`
- Code of the mini-case: [Quantlets/Ch_04/ai_discovery_case.py](#)

Conclusions

- ▣ Markowitz is the right question with fragile answers: estimation error dominates
- ▣ Sample MV needs decades of data to beat $1/N$; out of sample it fails in every universe
- ▣ Shrinkage, constraints, Black–Litterman and risk-based rules make portfolios stable
- ▣ No rule beats $1/N$ significantly here; several lose after costs
- ▣ Evaluate out of sample, net of costs, with a test of the Sharpe difference

Links to Other Chapters

- ▣ **Chapter 3** (factor models): the tangency portfolio and the CAPM; factor covariance models
- ▣ **Chapters 5–6**: time-varying volatility and correlation for dynamic weights
- ▣ **Chapter 7**: portfolio Value-at-Risk and Expected Shortfall
- ▣ **Chapter 8**: backtesting without overfitting
- ▣ **Chapter 13**: machine learning for returns and portfolios

Further Study on Quantinar

Advanced courses on quantinar.com

- ▣  Hierarchical Clustering
- ▣  Minimum Spanning Tree
- ▣  Multivariate Statistical Analysis

Code and data

- ▣ All charts and tables: github.com/danpele/MFM/tree/main/Quantlets/Ch_04
- ▣ Risk-free rate: Kenneth French Data Library

References (1/10)

- ▣ Andrews, D.W.K. (1991). Heteroskedasticity and autocorrelation consistent covariance matrix estimation. *Econometrica*, 59(3), 817–858.
- ▣ Andrews, D.W.K., Monahan, J.C. (1992). An improved heteroskedasticity and autocorrelation consistent covariance matrix estimator. *Econometrica*, 60(4), 953–966.
- ▣ Bailey, D.H., López de Prado, M. (2014). The deflated Sharpe ratio: correcting for selection bias, backtest overfitting, and non-normality. *Journal of Portfolio Management*, 40(5), 94–107.
- ▣ Benjamini, Y., Hochberg, Y. (1995). Controlling the false discovery rate: a practical and powerful approach to multiple testing. *Journal of the Royal Statistical Society: Series B*, 57(1), 289–300.
- ▣ Best, M.J., Grauer, R.R. (1991). On the sensitivity of mean-variance-efficient portfolios to changes in asset means: some analytical and computational results. *Review of Financial Studies*, 4(2), 315–342.
- ▣ Black, F., Litterman, R. (1992). Global portfolio optimization. *Financial Analysts Journal*, 48(5), 28–43.

References (2/10)

- Bodnar, T., Parolya, N., Schmid, W. (2018). Estimation of the global minimum variance portfolio in high dimensions. *European Journal of Operational Research*, 266(1), 371–390.
- Brandt, M.W., Santa-Clara, P., Valkanov, R. (2009). Parametric portfolio policies: exploiting characteristics in the cross-section of equity returns. *Review of Financial Studies*, 22(9), 3411–3447.
- Britten-Jones, M. (1999). The sampling error in estimates of mean-variance efficient portfolio weights. *Journal of Finance*, 54(2), 655–671.
- Chopra, V.K., Ziemba, W.T. (1993). The effect of errors in means, variances, and covariances on optimal portfolio choice. *Journal of Portfolio Management*, 19(2), 6–11.
- Choueifaty, Y., Coignard, Y. (2008). Toward maximum diversification. *Journal of Portfolio Management*, 35(1), 40–51.
- DeMiguel, V., Garlappi, L., Nogales, F.J., Uppal, R. (2009). A generalized approach to portfolio optimization: improving performance by constraining portfolio norms. *Management Science*, 55(5), 798–812.

References (3/10)

- ▣ DeMiguel, V., Garlappi, L., Uppal, R. (2009). Optimal versus naive diversification: how inefficient is the $1/N$ portfolio strategy?. *Review of Financial Studies*, 22(5), 1915–1953.
- ▣ Dragotă, M., Șerbănescu, C.I., Pele, D.T. (2008). Portfolio diversification and market share analysis for Romanian insurance companies. In *Risk Management and Value*, World Scientific, 277–298.
- ▣ Fan, J., Liao, Y., Mincheva, M. (2013). Large covariance estimation by thresholding principal orthogonal complements. *Journal of the Royal Statistical Society: Series B*, 75(4), 603–680.
- ▣ Franke, J., Härdle, W.K., Hafner, C.M. (2019). *Statistics of Financial Markets: An Introduction*, 5th ed. Springer.
- ▣ Frazzini, A., Israel, R., Moskowitz, T.J. (2018). Trading costs. SSRN Working Paper 3229719.
- ▣ Frost, P.A., Savarino, J.E. (1988). For better performance: constrain portfolio weights. *Journal of Portfolio Management*, 15(1), 29–34.

References (4/10)

- ▣ Gârleanu, N., Pedersen, L.H. (2013). [Dynamic trading with predictable returns and transaction costs](#). *Journal of Finance*, 68(6), 2309–2340.
- ▣ Härdle, W.K., Simar, L. (2019). [Applied Multivariate Statistical Analysis](#), 5th ed. Springer.
- ▣ Harvey, C.R., Liu, Y. (2015). [Backtesting](#). *Journal of Portfolio Management*, 42(1), 13–28.
- ▣ He, G., Litterman, R. (2002). [The intuition behind Black-Litterman model portfolios](#). SSRN Working Paper 334304.
- ▣ Jagannathan, R., Ma, T. (2003). [Risk reduction in large portfolios: why imposing the wrong constraints helps](#). *Journal of Finance*, 58(4), 1651–1683.
- ▣ Jensen, T.I., Kelly, B., Malamud, S., Pedersen, L.H. (2026). [Machine learning and the implementable efficient frontier](#). *Review of Financial Studies*, 39(10), 3035–3078.

References (5/10)

- ▣ Jensen, T.I., Kelly, B., Pedersen, L.H. (2023). *Is there a replication crisis in finance?*. *Journal of Finance*, 78(5), 2465–2518.
- ▣ Jobson, J.D., Korkie, B.M. (1981). *Performance hypothesis testing with the Sharpe and Treynor measures*. *Journal of Finance*, 36(4), 889–908.
- ▣ Jorion, P. (1986). *Bayes-Stein estimation for portfolio analysis*. *Journal of Financial and Quantitative Analysis*, 21(3), 279–292.
- ▣ Kan, R., Wang, X., Zhou, G. (2022). *Optimal portfolio choice with estimation risk: no risk-free asset case*. *Management Science*, 68(3), 2047–2068.
- ▣ Kan, R., Zhou, G. (2007). *Optimal portfolio choice with parameter uncertainty*. *Journal of Financial and Quantitative Analysis*, 42(3), 621–656.
- ▣ Künsch, H.R. (1989). *The jackknife and the bootstrap for general stationary observations*. *Annals of Statistics*, 17(3), 1217–1241.

References (6/10)

- ▣ Laloux, L., Cizeau, P., Bouchaud, J.-P., Potters, M. (1999). Noise dressing of financial correlation matrices. *Physical Review Letters*, 83(7), 1467–1470.
- ▣ Ledoit, O., Wolf, M. (2003). Improved estimation of the covariance matrix of stock returns with an application to portfolio selection. *Journal of Empirical Finance*, 10(5), 603–621.
- ▣ Ledoit, O., Wolf, M. (2004a). A well-conditioned estimator for large-dimensional covariance matrices. *Journal of Multivariate Analysis*, 88(2), 365–411.
- ▣ Ledoit, O., Wolf, M. (2004b). Honey, I shrunk the sample covariance matrix. *Journal of Portfolio Management*, 30(4), 110–119.
- ▣ Ledoit, O., Wolf, M. (2008). Robust performance hypothesis testing with the Sharpe ratio. *Journal of Empirical Finance*, 15(5), 850–859.
- ▣ Ledoit, O., Wolf, M. (2017). Nonlinear shrinkage of the covariance matrix for portfolio selection: Markowitz meets Goldilocks. *Review of Financial Studies*, 30(12), 4349–4388.

References (7/10)

- ▣ Ledoit, O., Wolf, M. (2020). Analytical nonlinear shrinkage of large-dimensional covariance matrices. *Annals of Statistics*, 48(5), 3043–3065.
- ▣ Lo, A.W. (2002). The statistics of Sharpe ratios. *Financial Analysts Journal*, 58(4), 36–52.
- ▣ López de Prado, M. (2016). Building diversified portfolios that outperform out of sample. *Journal of Portfolio Management*, 42(4), 59–69.
- ▣ Maillard, S., Roncalli, T., Teïletche, J. (2010). The properties of equally weighted risk contribution portfolios. *Journal of Portfolio Management*, 36(4), 60–70.
- ▣ Mantegna, R.N. (1999). Hierarchical structure in financial markets. *European Physical Journal B*, 11(1), 193–197.
- ▣ Marčenko, V.A., Pastur, L.A. (1967). Distribution of eigenvalues for some sets of random matrices. *Mathematics of the USSR-Sbornik*, 1(4), 457–483.

References (8/10)

- ▣ Markowitz, H. (1952). Portfolio selection. *Journal of Finance*, 7(1), 77–91.
- ▣ Merton, R.C. (1972). An analytic derivation of the efficient portfolio frontier. *Journal of Financial and Quantitative Analysis*, 7(4), 1851–1872.
- ▣ Merton, R.C. (1980). On estimating the expected return on the market: an exploratory investigation. *Journal of Financial Economics*, 8(4), 323–361.
- ▣ Michaud, R.O. (1989). The Markowitz optimization enigma: is 'optimized' optimal?. *Financial Analysts Journal*, 45(1), 31–42.
- ▣ Okhrin, Y., Schmid, W. (2006). Distributional properties of portfolio weights. *Journal of Econometrics*, 134(1), 235–256.
- ▣ Pele, D.T. (2023). Robust Markowitz for crypto-assets. Quantinar course, Bucharest University of Economic Studies.

References (9/10)

- ▣ Pele, D.T., Wesselhöfft, N., Härdle, W.K., Kolossiatis, M., Yatracos, Y.G. (2023). *Are cryptos becoming alternative assets?*. *The European Journal of Finance*, 29(10), 1064–1105.
- ▣ Petukhina, A., Trimborn, S., Härdle, W.K., Elendner, H. (2021). *Investing with cryptocurrencies – evaluating their potential for portfolio allocation strategies*. *Quantitative Finance*, 21(11), 1825–1853.
- ▣ Politis, D.N., Romano, J.P. (1994). *The stationary bootstrap*. *Journal of the American Statistical Association*, 89(428), 1303–1313.
- ▣ Sharpe, W.F. (1966). *Mutual fund performance*. *Journal of Business*, 39(S1), 119–138.
- ▣ Theil, H., Goldberger, A.S. (1961). *On pure and mixed statistical estimation in economics*. *International Economic Review*, 2(1), 65–78.
- ▣ Tobin, J. (1958). *Liquidity preference as behavior towards risk*. *Review of Economic Studies*, 25(2), 65–86.

References (10/10)

- ▣ Tu, J., Zhou, G. (2011). Markowitz meets Talmud: a combination of sophisticated and naive diversification strategies. *Journal of Financial Economics*, 99(1), 204–215.
- ▣ Data: Kenneth R. French Data Library, mba.tuck.dartmouth.edu/pages/faculty/ken.french/data_library.html.